GP-RareLink · deep dive for collaborators

Reading the rare-disease literature with a language model, and checking its work

Interim report, version 2, in about 45 minutes · 2026-11-09

Rare Genomics Institute with the Stanford Open Virtual Assistant Lab · funded by Anthropic's AI for Science program · internal review

What this talk covers

- Part 1. The problem, the question, and the five pre-registered outcomes
- Part 2. Methods: corpus, extraction, two checks on every value, records, the firewall and the harness
- Part 3. Results, in the report's order: corpus, gatekeeper, stability, records, the unrated audit, ClinVar, OMIM, gap candidates, bias and memorization, the open-weight model, exploratory analyses, each disease, spend
- Part 4. What it means, the gaps and a costed plan, and what we need from you
- Everything here is on the dashboard: the papers, this deck with its notes, the candidate list and the input form

Part 1. The problem and the question

A variant published once, and then lost

- Alexander disease: 182 causative variants in 550 published patients (2024 review) against 76 pathogenic or likely-pathogenic ClinVar records for GFAP (release 2026-10-12)
- Knowledge sits in ClinVar and OMIM for variants, HPO for phenotype vocabulary, Monarch's phenopacket store for a few cohorts, and in case reports for most individual patients
- A lab querying ClinVar today cannot see a causative variant published in 2011 that was never submitted; that family's diagnostic odyssey continues
- Too many diseases to curate by hand; the need recurs every time a team meets a disease nobody has curated
- The question: can the SLIDERS method, powered by Claude, build variant-phenotype-disease records from case reports at an accuracy we can prove, at a scale that materially closes the gap?

Three design choices that make the answer checkable

1 Quote everything

A value with no verbatim quote is not allowed. A script looks up every quote in the paper. "Not reported" is a first-class answer; three-state phenotypes (observed, excluded, not reported) are never collapsed.

2 Hold out the answer key

ClinVar and OMIM are never shown to the model. Code that reads papers cannot read reference data; an automated test fails if it could. Comparisons run afterwards in a separate harness.

3 Decide the rules first

Prompts, forms and the analysis plan were frozen on 2026-10-20 after a one-disease pilot (tags prereg-v1.0 and pipeline-v1.0). Changes since are numbered deviations approved by the PI.

Five pre-registered outcomes, and where each stands

#	Outcome	Type	Pre-specified target	Status
O1	Unsupported-value rate, human audit of 250 claims	primary, confirmatory	under 2%: at most 4 of 250 (rule C)	sample drawn, pages built; no claim rated
O2	Concordance with ClinVar and OMIM: gene, HGVS, classification, phenotype	primary, estimation	none; estimates with intervals	ClinVar done (6 diseases); OMIM variant half done 2026-10-28; OMIM phenotype half not assessed
O3	Gap closure: variants absent from both databases, adjudicated, submitted	primary, descriptive	counts by stage	214 candidates with dossiers; adjudication not started
O4	Ascertainment bias vs natural-history cohorts; memorization probe	secondary	none	probe done (both models); frequency deviation waits on a person
O5	Expert-minutes per record; Claude vs open-weight model	secondary	none	open-weight run and gatekeeper done (2 diseases); blinded audit and minutes not measured

Only O1 has a pass/fail rule. Headline pooled figures use the five diseases other than HA20, the development disease.

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Part 2. Methods

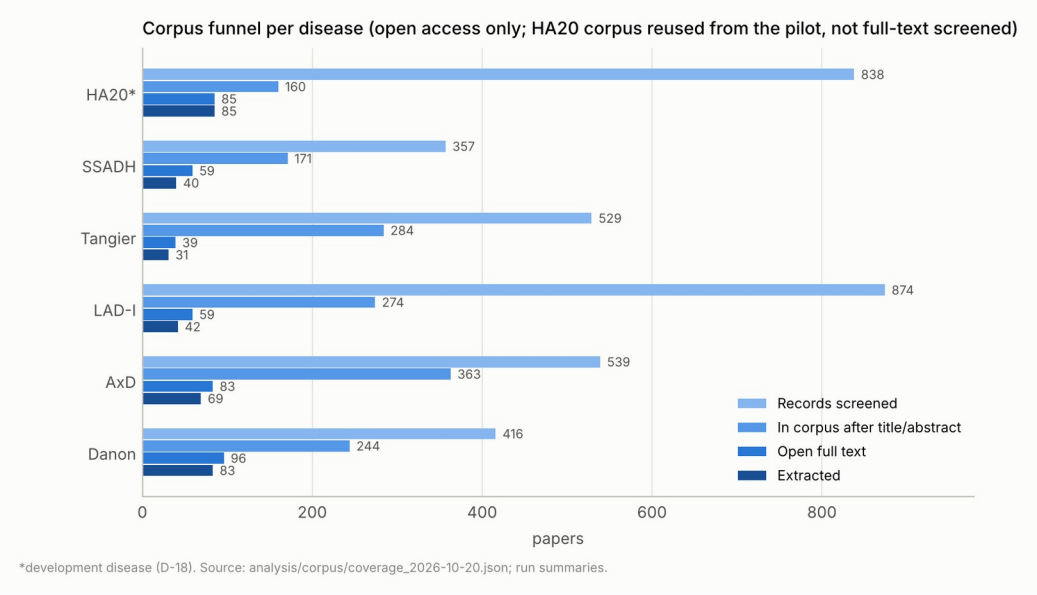
Corpus · extraction · two checks on every value · records · the firewall and the harness · outcomes and statistics · the pilot and the deviations

Six diseases, six organ systems, four inheritance modes

Disease	Gene	Inheritance	System	ClinVar records; P/LP; VUS (2026-10-12)	OMIM allelic variants (gene entry)
A20 haploinsufficiency (development)	TNFAIP3	AD	Auto-inflammatory	764; 99; 375	9
SSADH deficiency	ALDH5A1	AR	Neurometabolic	914; 152; 333	7
Tangier disease	ABCA1	AR	Lipid metabolism	2,006; 85; 705	19
Leukocyte adhesion deficiency I	ITGB2	AR	Immunology	949; 87; 345	14
Alexander disease	GFAP	AD / de novo	Neurology	627; 76; 262	13
Danon disease	LAMP2	X-linked dominant	Cardiac / neuromuscular	986; 154; 292	13

ClinVar: distinct VariationIDs on single-gene GRCh38 records; P/LP sums Pathogenic, Likely pathogenic and Pathogenic/Likely pathogenic. OMIM lists "selected examples", not an exhaustive catalogue.

Building the corpus



The corpus funnel per disease: screened, in corpus, open full text, extracted.

- Search: PubMed from inception, gene symbol plus disease synonyms, backward citation searching of key reviews; no ClinVar or OMIM seeds for the five new diseases (D-19)
- Screening: title/abstract by model helpers with a logged reason, then full-text screening with a verbatim quote checked by script; a 20-item human check (D-34)
- Eligibility: individual-level genotype or phenotype for at least one person; reviews with per-patient tables included and tagged (D-33)
- Access: open access only (D-32). 3,553 records screened, 1,496 in corpus, 421 with open full text
- Holdouts: natural-history comparator papers kept out of extraction (AxD 2, SSADH 1, Danon 2)

Extraction: one reader per paper, a quote per value

- Model and route: Claude Opus as Claude Code helper agents (claude-opus-5-5), one per paper; the helper sees only the paper, the instructions, the schema and the rules; no web access; no memory of the paper allowed
- Frozen files: extract_v6, schema_<disease>_v5 (publications, individuals, variants, phenotypic_features, phenotype_checklist), extraction_rules_v1, gatekeeper_v4; per-disease instantiation swaps only name, gene, schema path and checklist (D-38)
- What a value is: a verbatim quote for every asserted value; "not reported" is first class; three states never collapsed; variant fields only from statements about this person (D-29); no positional reading of stacked cells (D-31); a normal finding recorded as the abnormality excluded (D-27)
- Checklists: 20 disease-specific features with three-state status per individual, written from a guideline or review, every quote checked verbatim

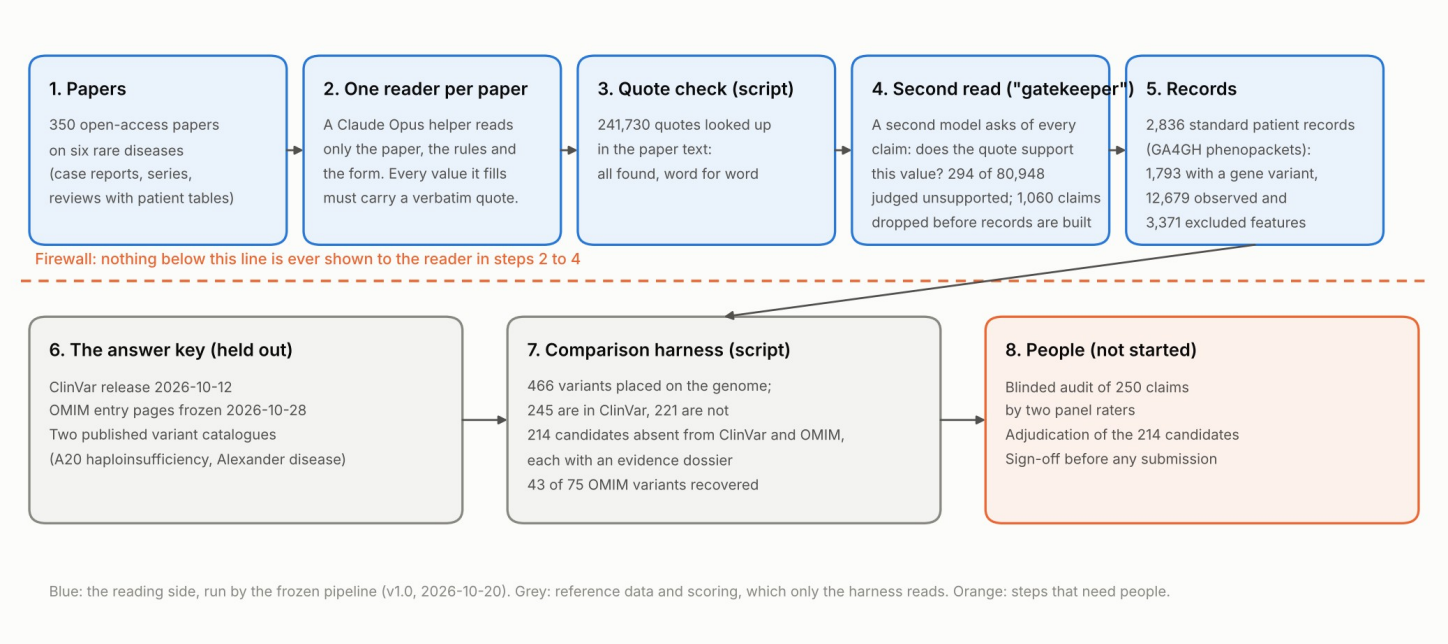
Two checks on every value: the span check and the gatekeeper

- Span check: every quoted span must occur verbatim in the source text; run on all values; repair-only resumes are diffed cell by cell
- Gatekeeper v4: a second Claude read of each model claim in chunks of up to 350, asking whether the quote entails the value for this person
- Records whom the quote is about: this person, a named group, the cohort in general, another person, the paper, unclear
- Four verdicts: supported, partially supported, unsupported, span not found. No script match exempts a value (D-28)
- Phenopackets keep only supported claims; the number removed is reported
- The gatekeeper is a filter, not a measure: in the pilot it understated the human audit by an order of magnitude

From rows to records

- Deduplication (merge policy v4): merge only on an explicit citation to an earlier report, one-to-one; a listed duplicate-publication pair (SSADH preprint and journal version) uses a dedicated route
- Normalization: VariantValidator 4.0.1, MANE Select transcript per gene; original string kept beside the normalized form; assumed or version-mapped transcripts flagged; only normalization fields enter, never database annotations
- Binding: checklist items to HPO through per-disease maps from the pinned release (2026-09-01); free-text features by script then helpers with a lookup tool; disease terms from Mondo (2026-10-06) by name
- Phenopackets: one GA4GH Phenopacket v2 per identity, schema-validated, per-disease settings (D-40)

The whole workflow, with the six-disease numbers



Blue: the reading side, run by the frozen pipeline. Grey: reference data and scoring, which only the harness reads. Orange: steps that need people.

The firewall and the harness

- An automated test fails if code under pipeline/extract can reach reference data, or even contains the tokens "clinvar", "omim" or "harness"
- Reference snapshots live outside the repository; only manifests and counts are committed
- ClinVar: variant_summary of the 2026-10-12 release, single-gene GRCh38 records per gene; the same release re-read on 2026-10-21 for the gap re-check
- HPO annotations (phenotype.hpoa, 2026-09-02) for the phenotype-profile figure
- Catalogues: Chen 2020 Table S1 for HA20 (D-30); Grossi 2024 Supplementary Table 2 for AxD (D-39); four other candidates failed the suitability check
- OMIM: twelve entry pages frozen by the PI on 2026-10-28 (D-42), parsed by the harness; allelic variants placed in ClinVar through ClinVar's OMIM cross-references; clinical synopses not frozen, so OMIM's phenotype half is "not yet assessed"

Outcomes and statistics, as registered

- O1: 250 claims, simple random sampling stratified by disease, seed 40250, drawn before any concordance; two blinded raters and a tie-breaker; pass if at most 4 unsupported (rule C); a value the rater cannot verify counts as unsupported
- O2: tier 1 = same GRCh38 allele; tier 2 = same protein change where no cDNA is given. HGVS recall = share of ClinVar's (PMID, variant) citation pairs in corpus papers that the run extracted. Overlap = share of extracted tier-1 alleles present in ClinVar (not precision). Classification = authors' call vs ClinVar aggregate
- O3: a gap candidate = tier-1 allele absent from the ClinVar snapshot, gatekeeper-supported in at least one row, and not among OMIM's allelic variants (D-42). Counted by stage: candidates, adjudicated, approved, drafted, signed off, submitted, accepted
- O4: a person lists each held-out comparator's HPO-mappable features and cohort frequencies first; then observed ÷ (observed + excluded) per feature with a publication-cluster bootstrap. Memorization: closed-book recall and guided completion on 30 papers per disease, stratified by date against the training cutoff
- O5: expert-minutes per accepted record (logged by the adjudication page); Claude vs DeepSeek-V4.1-Flash on identical papers, prompts, frozen files and scoring
- Intervals: Wilson for simple proportions; exact Clopper-Pearson for the audit and the gatekeeper shares

The development pilot, and four deviations since the freeze

- Pilot on HA20, 2026-10-18 to 2026-10-20: two full runs, a 15-paper repeat, every merge hand-checked, two 50-claim human audits (6.0% and 6.3% unsupported), ClinVar concordance, Chen 2020 completeness. HA20 then re-extracted from scratch with the frozen pipeline
- D-38 (2026-10-20): frozen prompt and schema instantiated per disease by script (name, gene, schema path, checklist only)
- D-39 (2026-10-20): the AxD catalogue's supplement arrived after extraction; transcribed on the harness side only
- D-40 (2026-10-20): Stage 4 scripts read per-disease settings instead of HA20's hard-coded ones
- D-43 (2026-10-28): isoform-sensitive positions in normalization (LAMP2 exons 9a/9b/9c, GFAP exon 7A); found through the OMIM holdout; the Danon run re-normalized and one candidate withdrawn
- Decisions, not deviations: the SSADH preprint and journal version both stay with their patients merged; D-42 adopted the frozen OMIM pages as the OMIM reference

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Part 3. Results

Corpus · extraction and the gatekeeper · stability · records · the unrated audit · ClinVar · OMIM · gap candidates · bias and memorization · the open-weight model · exploratory analyses · each disease · spend

Corpus and open-access coverage by disease

Disease	Records screened	In corpus	Open full text	Eligible after full-text screen	Papers extracted
HA20*	838	160	85 (53%)	not screened (pilot corpus)	85
SSADH	357	171	59 (35%)	40	40
Tangier	529	284	39 (14%)	31	31
LAD-I	874	274	59 (22%)	42	42
AxD	539	363	83 (23%)	69	69
Danon	416	244	96 (39%)	83	83
All six	3,553	1,496	421 (28%)		350

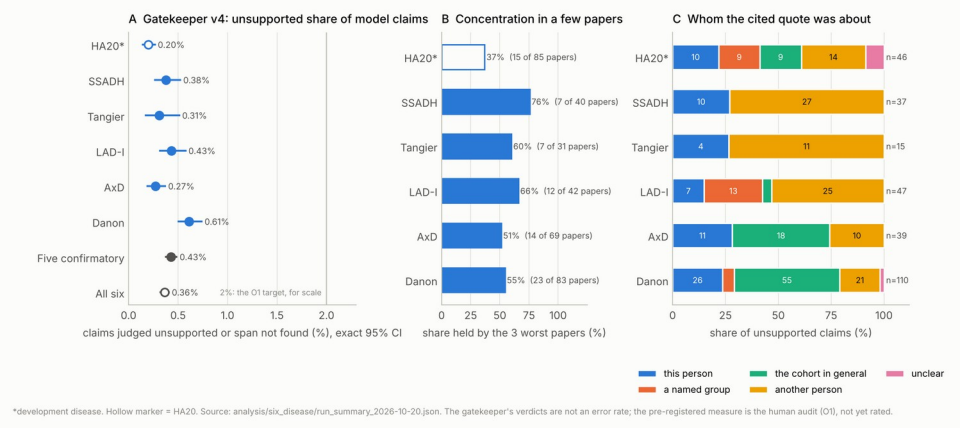
*HA20: development disease; its corpus was carried over from the pilot and kept its ClinVar-seeded papers, tagged (D-19). Open access only (D-32); no institutional access.

Extraction output and the gatekeeper's verdicts

Disease	Papers	Individual rows	Values with a quote	Span problems	Model claims judged	Unsupported	Not fully supported	Papers with any	Worst 3 papers' share
HA20*	85	811	67,863	0	23,129	0.20%	1.93%	15 of 85	37%
SSADH	40	421	31,370	0	9,759	0.38%	2.86%	7 of 40	76%
Tangier	31	235	16,762	0	4,830	0.31%	4.49%	7 of 31	60%
LAD-I	42	508	36,280	0	10,864	0.43%	3.47%	12 of 42	66%
AxD	69	463	40,773	0	14,403	0.27%	3.55%	14 of 69	51%
Danon	83	510	48,682	0	17,963	0.61%	3.07%	23 of 83	55%
Five confirmatory	265	2,137	173,867	0	57,819	0.43%	3.35%		
All six	350	2,948	241,730	0	80,948	0.36%	2.94%		

"Unsupported" counts unsupported and span-not-found verdicts; "not fully supported" adds partially supported. The gatekeeper is a filter, not a measure: this is not an error rate.

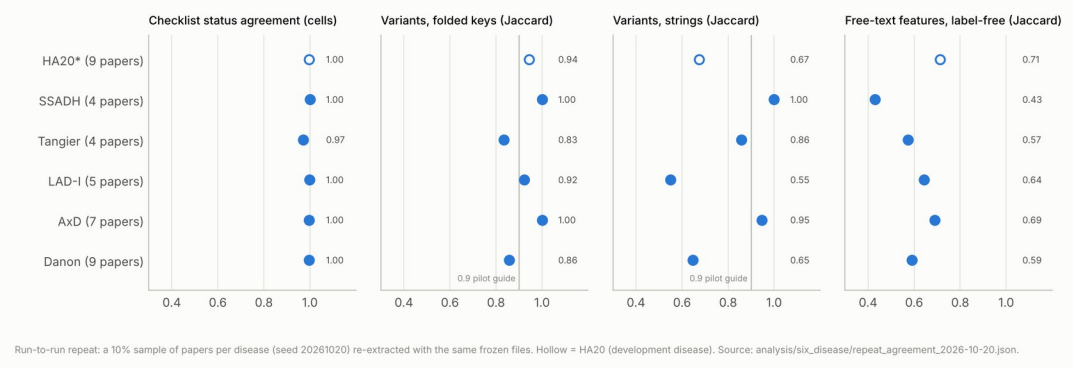
Why claims fail



A: unsupported share with exact intervals. B: share of each disease's unsupported claims held by its worst three papers. C: whom the cited quote was about, for unsupported claims.

- Commonest unsupported field in five diseases: the variant row's gene symbol, quoted from a sentence about the proband but applied to a relative or a cited patient (HA20 17, SSADH 16, Tangier 9, LAD-I 14, AxD 21 claims)
- Hedged wording ("suggested", "suspected", "consistent with", "typically") recorded as observed or excluded; extraction v6 has no hedge rule, known at the freeze
- Cohort-level statements applied to individuals: "the cohort in general" dominates Danon's and AxD's unsupported claims
- Danon's three worst papers hold 60 of 110: carriers given the proband's variant; "usually mild" applied to individuals; exclusions from "typically showed no symptoms"
- About 30 papers contradict themselves (table vs text); judged on the table, noted, collected for the human audit

Run-to-run stability



Agreement by measure. Checklist statuses are stable everywhere; variant keys clear 0.9 in four diseases; free-text features agree least.

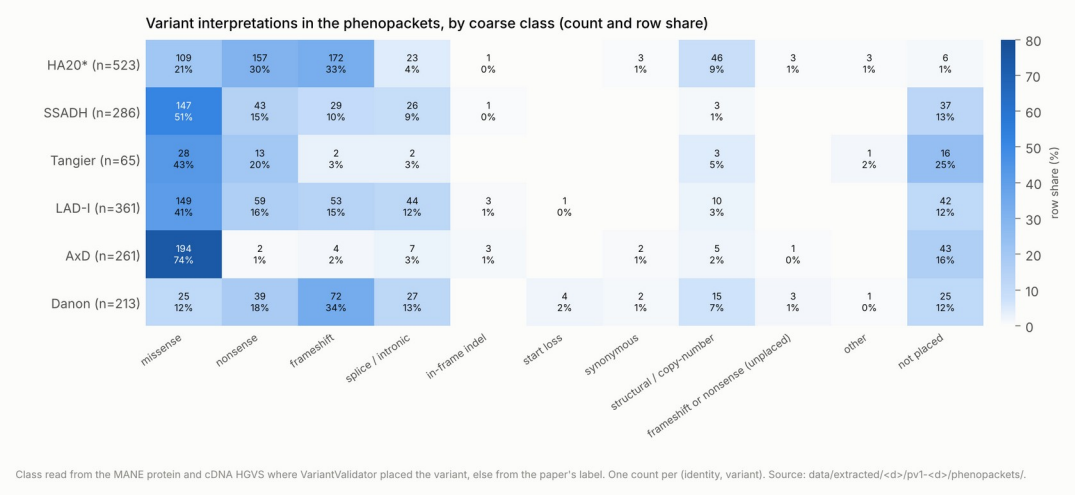
- 38 papers (10% per disease, seed 20261020) re-extracted with the same frozen files; gatekeeper not run on the repeat
- Checklist statuses: 97 to 100% of cells agree in every disease
- Variant keys: 83 to 100%; the dips (Tangier, Danon) are notation only (a full-width asterisk; "genetic deletion of exons 4 to 10" vs "deletion of exons 4 to 10")
- Same individual count in 31 of 38 papers; the differences are scope calls on relatives and on patients cited from earlier reports
- Free-text features agree least (label-free Jaccard 0.43 to 0.71): two readers phrase and split features differently

Records: from rows to phenopackets

Disease	Rows → identities (merges)	Review-only	With gene variant	cDNA normalized	Feature rows bound	Phenopackets (with interpretation)	Observed / excluded terms	Conflicts	Claims dropped
HA20*	811 → 769 (42)	282	558	97%; 85 protein-only	92%	769 (499)	3,678 / 1,018	93	207
SSADH	421 → 363 (58)	43	243	87%; 15 protein-only	90%	363 (174)	1,249 / 299	10	151
Tangier	235 → 235 (0)	140	65	90%; 17 protein-only	91%	235 (53)	745 / 219	13	117
LAD-I	508 → 508 (0)	194	328	86%; 7 protein-only	84%	508 (299)	1,264 / 299	5	100
AxD	463 → 456 (7)	175	318	79%; 65 protein-only	91%	456 (255)	2,441 / 529	40	221
Danon	510 → 505 (5)	164	281	89%; 25 protein-only	93%	505 (212)	3,302 / 1,007	48	264

All 2,836 phenopackets are schema-valid. "Review-only" identities come only from a review's per-patient table (D-33). Protein-only strings cannot be placed on the genome and match ClinVar only at tier 2.

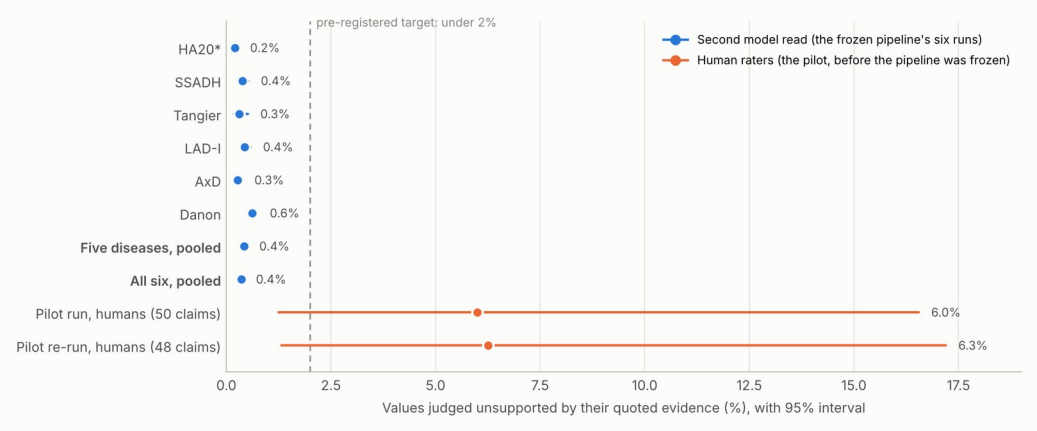
The variant spectrum matches expectation



Coarse variant classes per disease, from the normalized HGVS in the phenopackets.

- Missense dominates AxD (74%) and SSADH (51%); nonsense and frameshift dominate HA20 and Danon
- Most-observed features are the expected ones: HA20: Oral ulcer; SSADH: Global developmental delay; Tangier: Decreased circulating HDL-C concentration; LAD-I: Increased total leukocyte count; AxD: Abnormal cerebral white matter morphology; Danon: Left ventricular hypertrophy
- "Not reported" far outnumbers values for onset age and consanguinity, as the three-state design requires
- Sex unknown for HA20 175, SSADH 188, Tangier 28, LAD-I 233, AxD 147, Danon 41 identities, mostly from review tables
- Observed-excluded conflicts are few and kept visible in the records rather than resolved

O1 is not yet measured; what the pilot's audits showed



- O1: 250 claims drawn (seed 40250), blinded pages built for two raters and a tie-breaker; no claim rated
- Pilot audit 1 (run-2, 50 claims): 3 unsupported (6.0%), all variant fields from a group statement or summary table; gatekeeper v1 flagged none
- Pilot audit 2 (v5 re-run, 48 rated): 3 unsupported (6.3%), all ages read by position from stacked cells the rater could not verify
- Each audit produced a frozen rule (D-29: variant fields only from this person; D-31: no positional reading of stacked cells); none has been re-audited by a person
- The gatekeeper flagged 2 of 6 and 1 of 4 of the claims a person rejected, with no false flags: a filter, not a measure

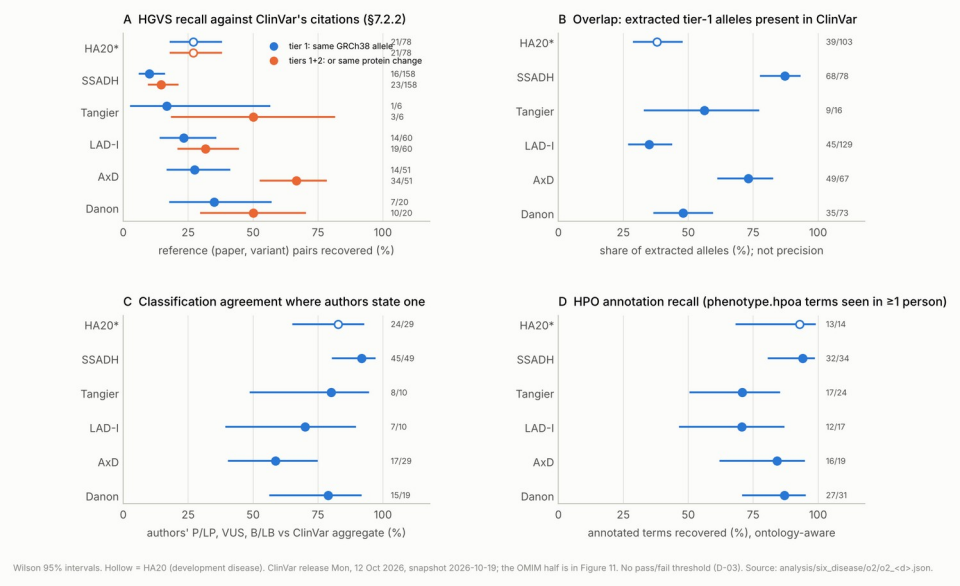
Blue: the gatekeeper's verdicts per disease run. Orange: two human audits of the pilot's output, before the fixes they produced were frozen in. Dashed: the 2% target.

O2: concordance with ClinVar, by disease

Disease	Gene matches	HGVS recall, tier 1	HGVS recall, tiers 1+2	Overlap, tier 1	Classification agreement	ClinVar condition names the disease	HPO annotation recall
HA20*	100% (558 of 559)	27% (21 of 78)	27% (21 of 78)	38% (39 of 103)	83% (24 of 29)	47% (27 of 57)	93% (13 of 14)
SSADH	100% (243 of 243)	10% (16 of 158)	15% (23 of 158)	87% (68 of 78)	92% (45 of 49)	99% (83 of 84)	94% (32 of 34)
Tangier	98% (65 of 66)	17% (1 of 6)	50% (3 of 6)	56% (9 of 16)	80% (8 of 10)	38% (6 of 16)	71% (17 of 24)
LAD-I	100% (328 of 328)	23% (14 of 60)	32% (19 of 60)	35% (45 of 129)	70% (7 of 10)	90% (47 of 52)	71% (12 of 17)
AxD	100% (318 of 319)	27% (14 of 51)	67% (34 of 51)	73% (49 of 67)	59% (17 of 29)	86% (95 of 111)	84% (16 of 19)
Danon	98% (281 of 287)	35% (7 of 20)	50% (10 of 20)	48% (35 of 73)	79% (15 of 19)	90% (43 of 48)	87% (27 of 31)

HGVS recall counts ClinVar's own (paper, variant) citation pairs restricted to corpus papers; overlap is not precision. Every miss counts as discordant until the registered human triage of 50 sampled misses is done.

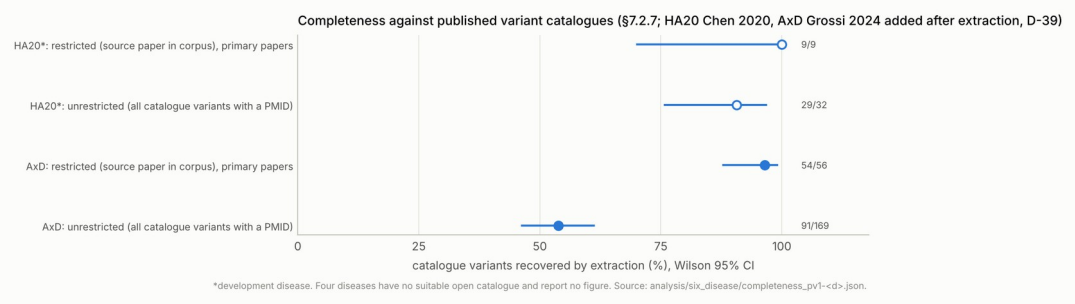
Reading the concordance figures



A: HGVS recall against ClinVar's citations. B: overlap of extracted alleles with ClinVar. C: classification agreement where authors state a class. D: HPO annotation recall.

- HGVS recall asks a narrow question: did the run extract, from this paper, the variant ClinVar cites to it? Misses concentrate in a few papers per disease
- SSADH: 68 misses in one paper (Akaboshi 2003, table scrambled in conversion) and most of the rest in Brennenstuhl 2020 (in-vitro variants, no patients)
- LAD-I: 37 of its misses in one review (van de Vijver 2012) whose "people" are accession numbers
- Overlap is not precision: an absent variant may be a real gap or a notation difference; the gap candidates below carry both possibilities until adjudicated
- Classification disagreement runs one way, as expected before registration: authors say P/LP, ClinVar says VUS
- A seeded sample of 50 misses awaits human triage into five categories (extraction, normalization, citation does not describe the variant, scope, other)

Completeness against two published catalogues



Completeness against the two published catalogues: restricted to variants whose source paper is in the corpus, and unrestricted.

- HA20 vs Chen 2020 Table S1 (D-30): 100% (9 of 9) restricted; 91% (29 of 32) unrestricted
- AxD vs Grossi 2024 Supplementary Table 2 (D-39): 96% (54 of 56) restricted; the two misses are notation; the catalogue's genomic column matched nothing as printed and was not used
- Restricted = catalogue variants whose source paper is in the corpus; unrestricted = all catalogue variants with a PMID. The gap between them is paywalled papers
- No registered catalogue for SSADH, Tangier, LAD-I or Danon (suitability check failed); finding one per disease is a cheap way to strengthen the study

The OMIM holdout: the source and the matching

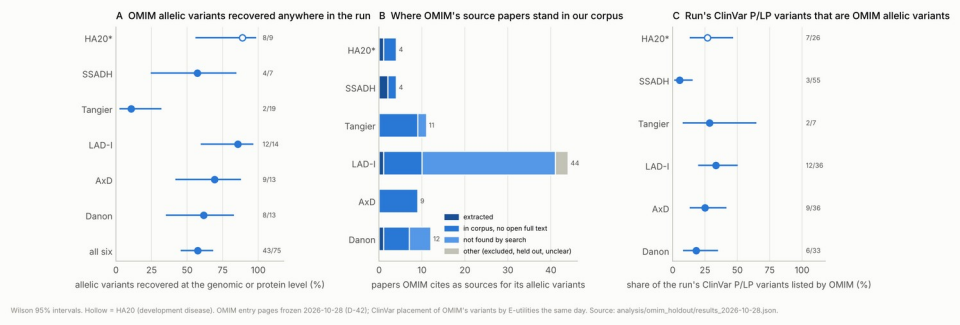
- Twelve omim.org entry pages frozen by the PI on 2026-10-28 (D-42); parsed by the harness; content kept outside the repository, only a checksummed manifest committed
- 82 allelic variants listed on the six gene entries; 75 titled for the six diseases; OMIM calls them "selected examples"
- Recall of OMIM's variants is meaningful; "precision against OMIM" is not; the gap definition's OMIM clause is one-sided
- Matching through ClinVar: OMIM cross-references and RCV accessions → ClinVar VariationID → current HGVS → match by allele, cDNA, protein, then a notation-tolerant token (reported apart)
- OMIM labels use legacy protein numbering (ABCA1 labels 60 residues below the current transcript; LAMP2 LEU113TER is p.Leu147Ter); dbSNP ids alone were not used because one rs covers every allele at a position
- Phenotype entries carry no allelic-variant table and no clinical synopsis; the synopsis pages are not frozen, so OMIM's phenotype half is "not yet assessed"

Recall of OMIM's allelic variants, by disease

Disease	OMIM allelic variants	Recovered (tiers 1+2)	Not recovered	Recall, Wilson 95%	OMIM source papers extracted	Corpus papers cited by OMIM
HA20*	9	8	1	89% (56 to 98)	1 of 4	3% (5 of 160)
SSADH	7	4	3	57% (25 to 84)	2 of 4	11% (18 of 171)
Tangier	19	2	17	11% (3 to 31)	0 of 11	12% (33 of 284)
LAD-I	14	12	2	86% (60 to 96)	1 of 44	5% (13 of 274)
AxD	13	9	4	69% (42 to 87)	0 of 9	6% (21 of 363)
Danon	13	8	5	62% (36 to 82)	1 of 12	6% (15 of 244)
All six	75	43	32	57% (46 to 68)	5 of 84	

Tier 1 = genomic allele or cDNA; tier 2 = protein change only. Token-level matches are not counted as recovered.

What the OMIM holdout taught



A: allelic variants recovered. B: where OMIM's source papers stand in our corpus funnel. C: the share of the run's ClinVar P/LP variants that OMIM lists.

- Of the 288 papers OMIM cites for these diseases: 9 extracted, 96 in corpus but paywalled, 160 never returned (mostly pre-1990 papers under earlier disease names)
- Where the source paper was read: 8 of 14 variants recovered from it; misses = a scrambled table and variants stated only in prose
- Patient level, 9 shared papers: 12 agree, 7 partial, 4 missed (all the scrambled table), none contradicted
- OMIM lists 75 selected examples; the run holds 466 genome-placed variants, and most of its ClinVar P/LP variants are not OMIM allelic variants
- OMIM is fallible: two allelic variants (610045.0002, 610045.0007) link to ClinVar records that contradict their text; 10 labels differ from ClinVar's protein change, 9 by legacy numbering

D-43: the candidate that was our own mistake

- All 215 candidates compared with OMIM's allelic variants by allele, cDNA, protein and notation token: 214 absent, 1 token collision
- G-danon-001 (NM_002294.3:c.1097_1098del, Sugie 2018): the paper's table says exon 9b (LAMP2B); the run had assumed the MANE transcript (exon 9a); the resulting allele exists in no database because it is not what the paper reported
- On the right transcript it is OMIM 309060.0001 and ClinVar 1410728: a normalization artefact, withdrawn (D-43)
- Fix: isoform-sensitive positions computed from RefSeq mRNA records (LAMP2 c.1094 onward; GFAP c.1172, exon 7A); the paper's exon or isoform label chooses the transcript, else no genomic key ("isoform_ambiguous")
- Danon re-normalized: 2 of 165 descriptions changed; phenopackets rebuilt (505, all valid); GFAP's one isoform case had already failed normalization safely
- Two independent references caught what one could not

O3: the gap, in numbers

466

distinct variants placed on the genome (GRCh38)

245

in ClinVar; 168 as pathogenic or likely pathogenic

221

absent from ClinVar in two successive releases

214

candidates after the gatekeeper, the OMIM check and one withdrawal, each with a dossier

4,725

quotes in the dossiers, all found verbatim

53 (15 strong)

in ClinVar only as VUS or conflicting while papers say pathogenic

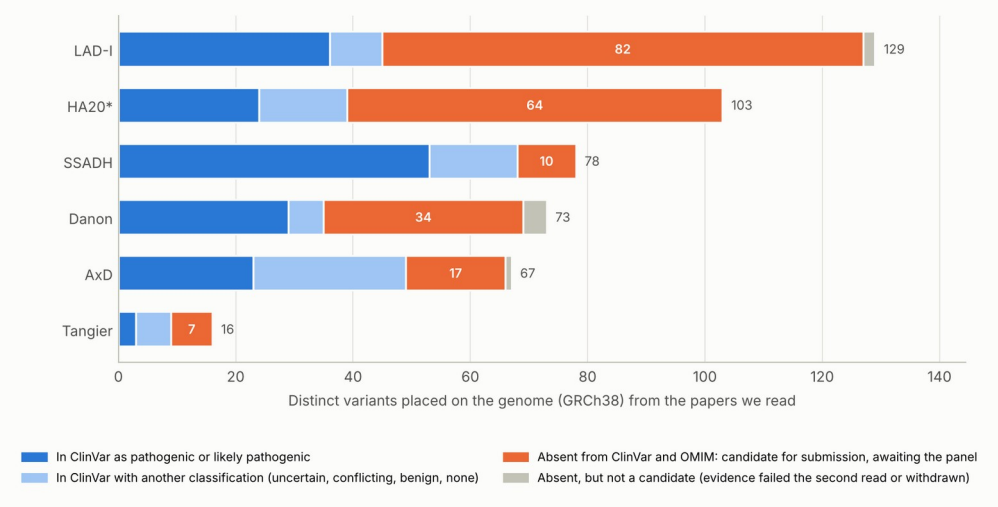
Candidate = tier-1 allele absent from the ClinVar snapshot, gatekeeper-supported in at least one row, not among OMIM's allelic variants. None adjudicated.

Gap candidates by disease, and what the flags say

Disease	Tier-1 alleles	Candidates (live)	In one paper only	Only from reviews	Only from ClinVar-cited papers	Protein check differs	Reclassification candidates (strong)
HA20*	103	64	43	0	6	8	12 (6)
SSADH	78	10	6	0	2	1	15 (4)
Tangier	16	7	7	0	1	1	6 (0)
LAD-I	129	82	76	34	66	10	7 (2)
AxD	67	17	15	1	2	3	8 (2)
Danon	73	34	27	0	7	2	5 (1)
All six	466	214	174	35	84	25	53 (15)

Triage flags (model-written, not verified): possible same person 77, other variant 51, carriers not attached 39, review only 35, protein mismatch 25, notation conflict 14, mosaic or somatic 13, possible existing record 1; 152 of 215 candidates carry at least one.

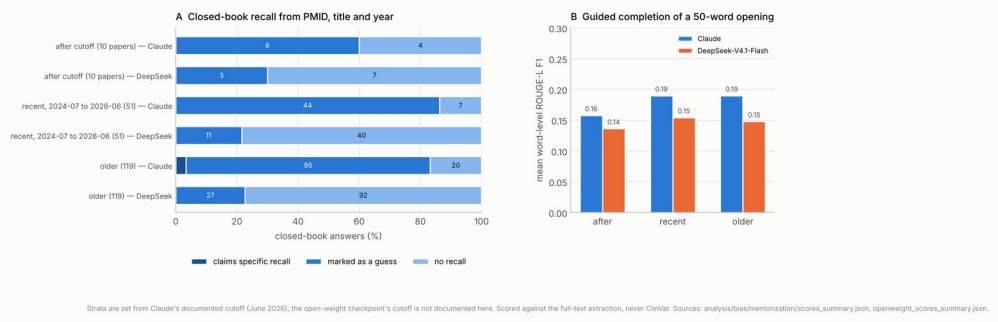
Where the found variants stand, and what a candidate looks like



- Dark blue: in ClinVar as P/LP. Light blue: in ClinVar with another classification. Orange: candidates. Grey: absent but not a candidate (evidence failed the second read, or withdrawn)
- A dossier per candidate: normalized description, the papers' own strings, every quote (verbatim-checked), individuals, zygosity where stated, authors' classification, flags
- The dashboard lists every candidate with PubMed links; the private adjudication page is ready for the panel
- Lower bound: open access only. Not yet gaps: adjudication sorts real gaps from notation, duplicates and carriers
- Helpers also proposed 19 same-person groups across the dossiers, not merged; if confirmed, up to a few percent of identities would collapse

Each bar: one disease's genome-placed variants. Orange: candidates awaiting the panel.

O4: the memorization probe



- 180 papers (30 per disease) by date vs the June 2026 cutoff: 10 after, 51 recent, 119 older; closed-book recall and guided completion; scored against the extraction, never ClinVar
- After cutoff: Claude 0 specific recalls, 0 variant hits; DeepSeek 0 and 0
- All strata: Claude claimed specific recall for 4 of 180 papers (all older, phenotypes only); variant hits were recurrent variants labelled guesses
- Completion: longest matching run 11 words, in formulaic consent sentences; mean 5-gram share under 1%
- Small after-cutoff stratum; black-box probes cannot see memorization that never surfaces as recall

Memorization probe on both models: neither reproduces the papers' variants or text; differences between strata are small and in no consistent direction.

O4: the frequency comparison has not started

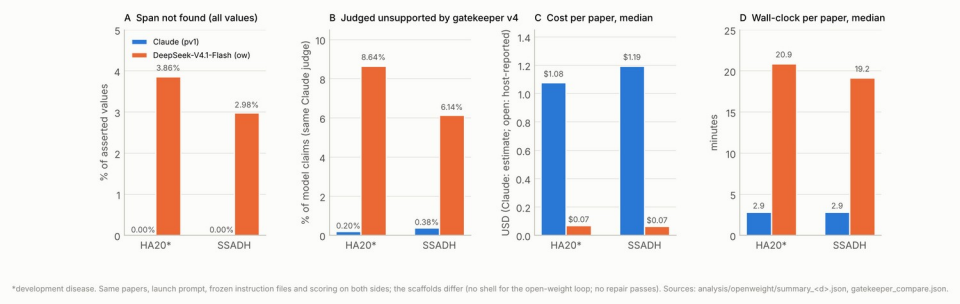
- Design: a person lists each comparator's HPO-mappable features and cohort frequencies before any literature frequency is computed
- Then observed ÷ (observed + excluded) per feature in our records, with a 2,000-resample publication-cluster bootstrap (seed 7425)
- Comparators fetched, with listing sheets: axd: gagliardi2024_axd_ukb; danon: khedro2025_danon_prospective; ha20: he2026_ha20_multicenter; ssadh: dibacco2019_ssadh_natural_history
- Three candidate comparators are paywalled (D-32); LAD-I and Tangier have no comparator and are out of the bias arm (D-35)
- Script written and tested on synthetic input only; needs 1 to 2 hours of a person per sheet

O5: Claude against an open-weight model on identical inputs

Disease	Papers	Values asserted (Claude / open)	Values with no quote	Judged unsupported	...among claims with a quote	Variant keys shared / union	Checklist agreement	Cost per paper, median	Minutes per paper, median
HA20*	85	67,863 / 55,317	0 / 2,003 (3.6%)	0.20% / 8.64%	0.20% / 2.28%	261 / 294 (89%)	95.9%	$1.08 / $0.07	3 / 21
SSADH	40	31,370 / 33,521	0 / 991 (3.0%)	0.38% / 6.14%	0.38% / 2.03%	157 / 166 (95%)	97.6%	$1.19 / $0.07	3 / 19

DeepSeek-V4.1-Flash (2026-09-10 checkpoint) on OpenRouter, provider pinned to DeepInfra fp8; same launch prompt, frozen files, file tools and scoring. Claude's cost is an estimate from helper tokens; the open model's is the host's reported price for final attempts.

Reading the comparison, and its cautions



Claude vs the open-weight model on the registered metrics measured so far.

- Variants: largely the same set found; HGVS recall against ClinVar overlaps within the intervals
- Provenance: Claude left no value without a quote; the open model left 3.6% and 3.0% without one, and some cells broke the schema shape
- Support, as the same Claude gatekeeper reads it: most of the gap is claims with no quote, but claims with a quote were also judged unsupported several times as often
- Cautions: the judge is Claude; helpers did not all treat empty quotes alike (236 HA20 and 144 SSADH empty-quote claims were judged supported against the row's quote)
- Cost and time: about 1/15 of the cost per paper, 7× the wall-clock, with waits after dropped connections (about 50 retries per 125 papers)
- Scaffold differences stated: no shell, no repair passes, three logged harness fixes during the run. Two diseases, one model, one provider

Exploratory: genotype-phenotype tests with GPSEA

Disease	Comparison	Individuals	Terms tested	Significant (BH 0.05)	Significant terms
HA20*	truncating vs missense	217	11	0	—
SSADH	biallelic truncating	77	9	0	—
Tangier	biallelic truncating	16	0	0	—
LAD-I	biallelic truncating	38	3	0	—
AxD	hotspot R79 R88 R239 vs other	83	14	0	—
Danon	truncating vs missense	130	9	2	Arrhythmia (57/59); Ventricular arrhythmia (45/47)
Danon	sex	130	10	3	Proximal muscle weakness (); Left ventricular hypertrophy (); Hypertrophic cardiomyopathy ()

Fisher exact tests, Benjamini-Hochberg at 0.05 (GPSEA 0.10.0). Zygosity for untyped calls is an analysis assumption; a feature not reported is not counted as excluded. Exploratory.

Exploratory: Tangier in ClinVar; deduplication limits

- ClinVar ABCA1: 2,006 single-gene records, 85 P/LP; 16 records (0.8%) and 7 P/LP records (8.2%) match the Tangier literature
- 19 of 35 extracted Tangier variants are not in ClinVar; most ABCA1 records come from unrelated testing
- LAD-I deduplication benchmark not computable: the review's supplement names no case pairs
- No citation merges in Tangier or LAD-I: a limit of the citation route; 19 proposed same-person groups await a person
- Both analyses exploratory (plan §7.6); the Tangier figure is a lower bound (open access only)

A20 haploinsufficiency (TNFAIP3) — development disease

- Corpus: 838 screened; 160 in corpus; 85 open full text (53%); 85 papers extracted. ClinVar (2026-10-12): 764 records, 99 P/LP
- Records: 811 rows → 769 identities (42 merges); 558 with a TNFAIP3 variant; 769 phenopackets, 499 with an interpretation; most observed: Oral ulcer 43%; Recurrent fever 27%; Inflammatory abnormality of the skin 22%
- Variants: 280 distinct descriptions; cDNA normalized 97%; classes frameshift 33%, nonsense 30%, missense 21%
- Gatekeeper: 0.20% unsupported (46 of 23,129); 15 of 85 papers. Repeat (9 papers): checklist 99.6%, variant keys 94%
- ClinVar: HGVS recall tier 1 27% (21 of 78), tiers 1+2 27% (21 of 78); overlap 38% (39 of 103); classification agreement 83% (24 of 29)
- Gaps and OMIM: 64 live candidates of 103 placed alleles; 12 reclassification candidates; OMIM 8 of 9 recovered, 1 of 4 OMIM source papers extracted
- Development disease: every rule was tuned on its pilot runs; 92 of 160 corpus papers were ClinVar-seeded (D-19). Treat its figures as an upper bound
- Completeness vs Chen 2020: 100% (9 of 9) restricted; most catalogue source papers are paywalled
- 12 proposed same-person groups, the most of any disease: HA20 reviews re-cite the same families

SSADH deficiency (ALDH5A1)

- Corpus: 357 screened; 171 in corpus; 59 open full text (35%); 40 papers extracted. ClinVar (2026-10-12): 914 records, 152 P/LP
- Records: 421 rows → 363 identities (58 merges); 243 with a ALDH5A1 variant; 363 phenopackets, 174 with an interpretation; most observed: Global developmental delay 25%; Seizure 20%; Hypotonia 18%
- Variants: 162 distinct descriptions; cDNA normalized 87%; classes missense 51%, nonsense 15%, not placed 13%
- Gatekeeper: 0.38% unsupported (37 of 9,759); 7 of 40 papers. Repeat (4 papers): checklist 100.0%, variant keys 100%
- ClinVar: HGVS recall tier 1 10% (16 of 158), tiers 1+2 15% (23 of 158); overlap 87% (68 of 78); classification agreement 92% (45 of 49)
- Gaps and OMIM: 10 live candidates of 78 placed alleles; 15 reclassification candidates; OMIM 4 of 7 recovered, 2 of 4 OMIM source papers extracted
- Duplicate study: the preprint and journal version are both in the corpus, their patients merged with evidence recorded; the papers disagree on G176R vs G176E for one variant
- O2 misses are two papers: a table scrambled in conversion (about 12 of 54 families extracted) and an in-vitro study with no patients
- Highest overlap with ClinVar of the six; only 10 candidates: a well-covered gene

Tangier disease (ABCA1)

- Corpus: 529 screened; 284 in corpus; 39 open full text (14%); 31 papers extracted. ClinVar (2026-10-12): 2,006 records, 85 P/LP
- Records: 235 rows → 235 identities (0 merges); 65 with a ABCA1 variant; 235 phenopackets, 53 with an interpretation; most observed: Decreased circulating HDL-C concentration 21%; Angina pectoris 14%; Coronary artery atherosclerosis 13%
- Variants: 48 distinct descriptions; cDNA normalized 90%; classes missense 43%, not placed 25%, nonsense 20%
- Gatekeeper: 0.31% unsupported (15 of 4,830); 7 of 31 papers. Repeat (4 papers): checklist 97.0%, variant keys 83%
- ClinVar: HGVS recall tier 1 17% (1 of 6), tiers 1+2 50% (3 of 6); overlap 56% (9 of 16); classification agreement 80% (8 of 10)
- Gaps and OMIM: 7 live candidates of 16 placed alleles; 6 reclassification candidates; OMIM 2 of 19 recovered, 0 of 11 OMIM source papers extracted
- Smallest corpus and lowest open-access share; many old papers with tables as images
- Only 65 of 235 identities carry an ABCA1 variant: 140 come from review tables and many older reports give no genotype
- ClinVar ABCA1 is dominated by unrelated testing; no comparator, no catalogue; GPSEA could not run

Leukocyte adhesion deficiency I (ITGB2)

- Corpus: 874 screened; 274 in corpus; 59 open full text (22%); 42 papers extracted. ClinVar (2026-10-12): 949 records, 87 P/LP
- Records: 508 rows → 508 identities (0 merges); 328 with a ITGB2 variant; 508 phenopackets, 299 with an interpretation; most observed: Increased total leukocyte count 32%; Neonatal omphalitis 12%; Pyoderma gangrenosum 11%
- Variants: 192 distinct descriptions; cDNA normalized 86%; classes missense 41%, nonsense 16%, frameshift 15%
- Gatekeeper: 0.43% unsupported (47 of 10,864); 12 of 42 papers. Repeat (5 papers): checklist 99.8%, variant keys 92%
- ClinVar: HGVS recall tier 1 23% (14 of 60), tiers 1+2 32% (19 of 60); overlap 35% (45 of 129); classification agreement 70% (7 of 10)
- Gaps and OMIM: 82 live candidates of 129 placed alleles; 7 reclassification candidates; OMIM 12 of 14 recovered, 1 of 44 OMIM source papers extracted
- The gap list is dominated by one review (van de Vijver 2012) whose people are accession numbers; many candidates may be numbering slips
- Input problem: one supplement cut at 60,000 characters, so patients P75 to P127 have no clinical rows
- Gene-name spelling (ITGβ2, ITGß2, CD18) covers 148 variant rows; the matcher accepts them

Alexander disease (GFAP)

- Corpus: 539 screened; 363 in corpus; 83 open full text (23%); 69 papers extracted. ClinVar (2026-10-12): 627 records, 76 P/LP
- Records: 463 rows → 456 identities (7 merges); 318 with a GFAP variant; 456 phenopackets, 255 with an interpretation; most observed: Abnormal cerebral white matter morphology 19%; Seizure 16%; Dysarthria 14%
- Variants: 181 distinct descriptions; cDNA normalized 79%; classes missense 74%, not placed 16%, splice / intronic 3%
- Gatekeeper: 0.27% unsupported (39 of 14,403); 14 of 69 papers. Repeat (7 papers): checklist 99.7%, variant keys 100%
- ClinVar: HGVS recall tier 1 27% (14 of 51), tiers 1+2 67% (34 of 51); overlap 73% (49 of 67); classification agreement 59% (17 of 29)
- Gaps and OMIM: 17 live candidates of 67 placed alleles; 8 reclassification candidates; OMIM 9 of 13 recovered, 0 of 9 OMIM source papers extracted
- Lowest normalization (79%): cDNA printed without "c.", legacy numbering; 65 protein-only strings match ClinVar only at tier 2
- Completeness vs Grossi 2024: 96% (54 of 56) restricted; the two misses are notation
- Missense dominates, as expected for a gain-of-function leukodystrophy; GPSEA found no hotspot difference

Danon disease (LAMP2)

- Corpus: 416 screened; 244 in corpus; 96 open full text (39%); 83 papers extracted. ClinVar (2026-10-12): 986 records, 154 P/LP
- Records: 510 rows → 505 identities (5 merges); 281 with a LAMP2 variant; 505 phenopackets, 212 with an interpretation; most observed: Left ventricular hypertrophy 39%; Hypertrophic cardiomyopathy 28%; Congestive heart failure 20%
- Variants: 165 distinct descriptions; cDNA normalized 89%; classes frameshift 34%, nonsense 18%, splice / intronic 13%
- Gatekeeper: 0.61% unsupported (110 of 17,963); 23 of 83 papers. Repeat (9 papers): checklist 99.6%, variant keys 86%
- ClinVar: HGVS recall tier 1 35% (7 of 20), tiers 1+2 50% (10 of 20); overlap 48% (35 of 73); classification agreement 79% (15 of 19)
- Gaps and OMIM: 34 live candidates of 73 placed alleles; 5 reclassification candidates; OMIM 8 of 13 recovered, 1 of 12 OMIM source papers extracted
- Highest gatekeeper unsupported share (0.61%): three papers hold 60 of 110
- Sex difference reproduced (GPSEA): males more often show myopathy and hypertrophy; a reproduction check
- D-43: one candidate withdrawn as an isoform artefact; LAMP2B-numbered variants need the panel's attention

Spend by budget line

Line	Budget line	Requested	Spent (estimate)	Share
1	Extraction across six diseases, incl. re-runs as HGVS and negation handling stabilize	$8,000	$1,132.12	14.2%
2	Gatekeeper verification pass	$4,000	$540.07	13.5%
3	ClinVar/OMIM reconciliation + concordance harness (full re-runs)	$6,000	$64.10	1.1%
4	Claude Code: pipeline, VariantValidator integration, dedup tooling, adjudication interface	$6,000	$15.46	0.3%
5	Ascertainment-bias arm	$4,000	$5.34	0.1%
6	Claude vs open-weight head-to-head	$2,000	$0.00	0.0%
7	Stretch: extend to all 12 candidate diseases	$9,000	$0.00	0.0%
8	Contingency	$6,000	$0.00	0.0%
—	Not yet assigned to a line		$103.97	
	Total, this session's readings	$45,000	$1,861.06	4.1%

Session usage readings at list price, including helper agents; not billing figures (D-15). Open-weight compute ($20) billed outside the award. An exploratory pilot in a separate session (about $317) is not covered here.

GP-RareLink deep dive

Part 4. What it means, the gaps, and what we need from you

What the data say, and what they do not

1 What the data say

End to end on six diseases at the projected scale (350 papers against "roughly 300"); provenance holds on every value; checklist repeats are stable; 2,836 valid phenopackets; intervals on every concordance figure; 214 candidates absent from two ClinVar releases and from OMIM. Three directional expectations stated before registration are consistent with what was seen.

2 What the data do not say

Nothing establishes the frozen pipeline's accuracy. The gatekeeper's 0.43% is a filter's reading; the human figures (6.0%, 6.3%) are from the pilot. OMIM recall (57%) measures coverage as much as extraction. Low ClinVar recall awaits triage. Two diseases, one open-weight model, one judge model.

3 What the gatekeeper is for

The records, not the metric: it removed 1,060 claims before the phenopackets were built, and its reasons name the failure modes to look for first (relatives' variant fields, hedged wording, cohort statements). O1 will give its accuracy against human labels, which is the number to report.

What the OMIM holdout adds

1 A direction

Where OMIM's source paper was read, the curated variant was almost always in our record with the right individuals and zygosity, with richer per-patient detail than OMIM's summary. Where it was not, the cause was access or search, not reading.

2 A correction

The OMIM check caught a candidate that was a transcript-isoform artefact of our normalization; the ClinVar check could not see it because the wrong allele was absent from ClinVar too. The fix (D-43) is in the pipeline.

3 A caution

OMIM carries legacy numbering and two allelic variants linked to the wrong ClinVar records. Concordance figures against OMIM should footnote them; a human-curated holdout needs the same provenance checks as a machine's output.

Gaps, part 1: what needs people

#	Gap	What it blocks	Who acts	Rough cost
G1	O1 human audit not rated (250 claims)	The primary outcome; gatekeeper accuracy vs human labels; any accuracy claim	PI shares the two rater pages; two panel members rate; a third breaks ties	8 to 25 panel hours; $0 model
G2	OMIM's phenotype half not assessed; OMIM read from frozen pages, not an API	Phenotype concordance against OMIM's clinical synopses; the OMIM submission route	PI freezes the six synopsis pages (or the API request lands); one harness run	$0 model; about a day of scripting
G3	214 gap candidates not adjudicated	Approved variants, submission drafts, expert-minutes, the gap-closure deliverable	PI shares the adjudication page; the panel reviews the dossiers; PI signs off	214 × 10 to 15 min ≈ 36 to 54 panel hours
G4	O2 miss triage not done (50 sampled misses)	Whether low ClinVar recall is extraction error, normalization, or citations that do not describe the variant	A panel member or the PI, on the private triage page	2 to 4 hours
G5	Blinded Claude vs open-weight audit not drawn	The registered O5 accuracy measure; the only neutral check on the gatekeeper's reading	Size with the panel's hours; draw by seed; same raters, blind to model	about 7 to 20 panel hours for n = 100
G6	Bias arm has no numbers	Per-disease frequency deviation; the ascertainment-bias deliverable	A person fills four comparator sheets without looking at literature frequencies; then the script runs	4 sheets × 1 to 2 hours

From the full report's section 5 (Table 16). Hours are guesses until the adjudication page logs real minutes.

Gaps, part 2: access, inputs, notation, bookkeeping

#	Gap	What it blocks	Who acts	Rough cost
G7	Open access only: 14 to 53% of each corpus readable	Completeness denominators; recall of the paywalled literature; the gap count is a lower bound	None within the rules; institutional access would be a deviation and a re-run	about $2 per paper plus gatekeeper; ~300 papers ≈ $600 to 900
G8	Extraction gaps frozen on purpose: no hedge rule; relatives' variant fields; cohort statements	A share of unsupported claims; would need a v7 and a deviation	Wait for O1; if it fails or shows these classes, propose v7 and re-run	full re-run ≈ $800 to 1,000
G9	Input conversion problems in about 10 papers (scrambled or image tables, truncated supplements)	Recall in those papers; several O2 misses	Fix the converter or refetch; re-extract under a logged deviation	$40 to 60 for 12 papers
G10	Notation and normalization: legacy numbering, protein-only strings, variants stated only in prose	Tier-1 matching: recall, overlap, candidates for those variants	Harness-side query corrections; a narrow rule for prose protein changes	$0 model; 1 to 2 days
G18	Historical disease names absent from the searches (38 of 84 OMIM source papers never returned)	Corpus recall of the founding literature for LAD-I, Danon and Tangier	Add the names to the search configuration and re-search; a deviation	$0 search; screening ≈ $0.05 per record
G11-G19	Same-person groups unmerged; HPO maps unreviewed; probe underpowered; minutes unmeasured; budget unreconciled; tags and OSF pending; drafts unsent	Deduplicated counts; HPO review; the public timestamp on the freeze; outreach	Mostly the PI; a few person-hours	hours

Items G8 to G10 and G18 change the frozen pipeline or corpus and should wait for the audit's measured failure rate.

A recommended order

Step	Do	Why first
1	Settle the panel (D-11) and share the O1 pages; draw the O5 audit sample at the same time (G1, G5)	Unblocks the primary outcome and the only neutral check on the gatekeeper
2	Freeze the OMIM synopsis pages or submit the API request (G2); push the tags; register on OSF; send the drafts (G16, G17)	No cost; long lead times start now
3	Triage the 50 O2 misses and LAD-I's 82 candidates first (G4, part of G3)	Tells us whether low recall and the biggest gap list are real before the panel spends hours on them
4	Fill the four comparator sheets (G6); review the checklist maps (G12); confirm the merge groups (G11)	Small human tasks that each unblock a registered figure
5	Only after O1 reads: decide on v7 and a re-run (G8), the input repairs (G9), the notation repairs (G10), the historical-name re-search (G18)	Changes to the frozen pipeline are deviations and should answer a measured failure, not an expected one
6	Panel adjudication of the remaining candidates (G3), then drafts and sign-off	Months 4 to 5 per plan; the dossiers and the page are ready

Steps 1 to 4 cost no model spend and about 60 to 110 panel hours in all, most of it adjudication. Nothing in steps 1 to 4 touches the frozen pipeline.

Limitations of this report

- No human-rated accuracy for the frozen pipeline: every support figure is a model's reading of a model's output
- HA20 is in every table, flagged; its rules were tuned on its own papers. Quote the five-disease figures
- OMIM is a sample, and only its variant half is in; the patient-level comparison was judged by Claude, not a panel member
- Open access only: 14 to 53% of the eligible literature; completeness and recall are measured on what was accessible
- Cost figures are session estimates with frozen readings and overlapping stages; the open-weight key is shared
- Helper-reported problems (input conversion, self-contradicting papers, likely non-gaps) are model output, not verified by a person
- The author of the report is the pipeline's author: the numbers are reproducible from the committed files; the framing is not independent of the system it describes

What we need from you

Role	What you would do	Effort	What it unlocks
Audit rater	Rate 250 extracted values against their quoted evidence on a blinded web page; two raters and a tie-breaker	10 to 15 hours per rater	The primary outcome: the first true accuracy figure
Adjudicator	Review the 214 candidate variants from their dossiers and decide whether the evidence supports a submission	10 to 15 minutes per candidate	ClinVar literature-only submissions and OMIM candidates
Cohort reader	Fill a comparator sheet of clinical-feature frequencies from a natural-history paper, without looking at our records	1 to 2 hours per sheet	The ascertainment-bias arm
Methods or domain reviewer	Read the short paper or the full report and tell us what is wrong, missing or unclear, from your field's point of view	1 to 3 hours	A better study before anything is submitted

Tell us who you are and how you would like to contribute through the form on the dashboard. Everything is published literature; there is no patient data.

Thank you

The dashboard holds the short paper, the full report and supplement, this deck, the videos, the candidate list and the form.

https://cjimmylin.github.io/gp-rarelink/

Every number in this talk comes from the project repository at commit 0e9390d. Still open: the human audit, the access decision, the adjudication of the candidates.